

Index Corruption – Assignment IV

DBMS Lab – Assignment IV

This document describes the group, schema design, and technology stack for the Student Academic Management System (DBMS Lab Assignment IV).

1 Group Name and Members

Group Name: Index Corruption

Members:

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2 Schema Design

The system uses a PostgreSQL database with the following tables. The schema supports universities, programs, courses, students, instructors, enrollments, teaching assignments, content items, and role-based access control.

2.1 RBAC and Users

Table	Description
app_user	Roles-based access control: id (PK), email (unique), password_hash, role (admin / instructor / student / analyst), created_at. Links to student.email or instructor.email.

2.2 Core Domain Tables

2.3 People and Assignments

2.4 Relationships (Summary)

- **Course** belongs to one University and one Program; has one Textbook; many ContentItems; many Enrollments; many TeachingAssignments; many Topics via course_topic.
- **Student** has many Enrollments; email links to app_user for login.
- **Instructor** has many TeachingAssignments; email links to app_user for login.
- **Enrollment** is a many-to-many between Student and Course with enroll_date and evaluation_score.
- **Teaching assignment** is a many-to-many between Instructor and Course with role.

Table	Key Columns
university	university_id (PK), name (unique), country
program	program_id (PK), program_name, program_type, duration_weeks_or_months
course	course_id (PK), course_name (unique), duration_weeks, university_id (FK), program_id (FK), textbook_id (FK)
textbook	textbook_id (PK), title, isbn (unique), url
topic	topic_id (PK), topic_name (unique)
course_topic	course_id (FK, PK), topic_id (FK, PK)

Table	Key Columns
student	student_id (PK), email (unique), full_name, age (≥ 13), country, category, skill_level
instructor	instructor_id (PK), full_name, email (unique)
enrollment	student_id (FK, PK), course_id (FK, PK), enroll_date, evaluation_score
teaching_assignment	instructor_id (FK, PK), course_id (FK, PK), role
content_item	content_id (PK), course_id (FK), content_type, title, url

3 Entity-Relationship Diagram

The complete ER diagram for the Student Academic Management System is shown below.

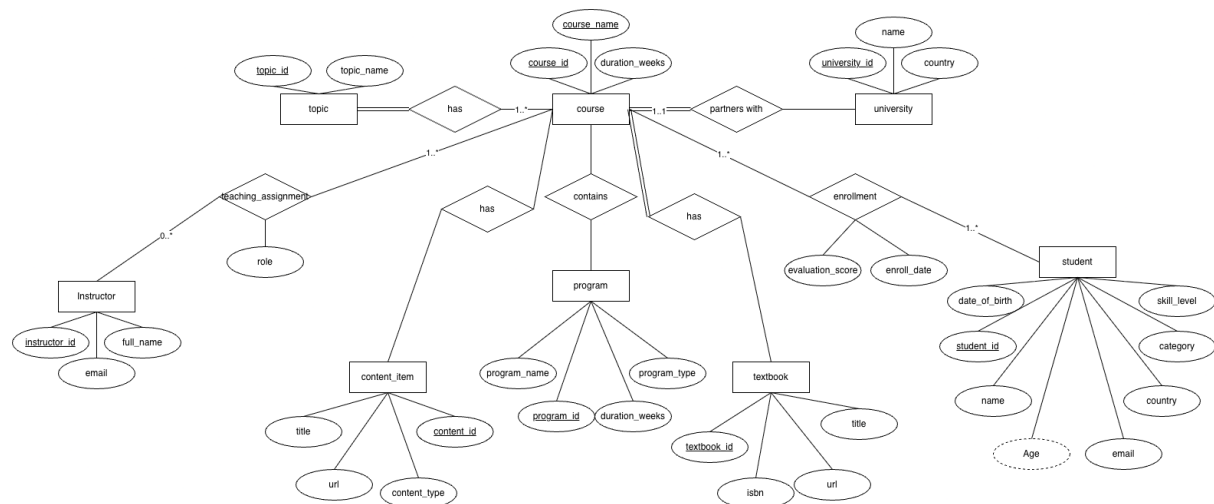


Figure 1: Entity-Relationship Diagram for Student Academic Management System

4 Languages and Technology Stack

4.1 Overview

- **Frontend:** Next.js (App Router), React, TypeScript
- **Backend:** FastAPI (Python 3.11+)
- **Database:** PostgreSQL
- **Build/Tooling:** Make, uv (or pip), npm

4.2 Backend (API)

- **Language:** Python 3.11+
- **Framework:** FastAPI
- **ORM / DB:** SQLAlchemy 2.x (async), asyncpg (PostgreSQL driver)
- **Validation & Settings:** Pydantic, pydantic-settings
- **Auth:** JWT (python-jose), password hashing (passlib/bcrypt)
- **Server:** Uvicorn

4.3 Frontend (Web)

- **Language:** TypeScript
- **Framework:** Next.js 16 (App Router)
- **UI:** React 19, Tailwind CSS 4, Radix UI (dialogs, labels, scroll-area, etc.), Lucide React (icons)
- **Linting:** ESLint, eslint-config-next

4.4 Project Structure

- `/apps/api` – FastAPI backend (routers: auth, student, instructor, admin, analyst)
- `/apps/web` – Next.js frontend (admin, analyst, instructor, student, login)
- `/docs` – Documentation (API.md, SCHEMA.md) and this submission document